ΠΑΝΕΠΙΣΤΗΜΙΟ ΚΡΗΤΗΣ
Τμήμα Επιστήμης Υπολογιστών

ΗΥ-112: Φυσική Ι
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Άσκηση 1. Έχουμε (1): $\vec{\alpha} = -2\hat{i} - 6\hat{j}$ και (2): $\vec{\beta} = -\hat{i} + 2\hat{j}$ διανύσματα. (Έστω $\vec{b} = \vec{\beta}$)!

(α) Είναι $\vec{\alpha} = (x_1, y_1) = (-2, -6)$ άρα $|\vec{\alpha}| = \sqrt{x_1^2 + y_1^2} = \sqrt{(-2)^2 + (-6)^2} = \sqrt{4+36} = \sqrt{40}$ μον.

και $\Theta_1 = \tan^{-1}\left(\frac{y_1}{x_1}\right) = \tan^{-1}\left(\frac{-6}{-2}\right) = \tan^{-1}(3) \simeq 71.57° \to +\pi = +180° = 251.57°$ (x)

↳ $71.57° - \pi \simeq 71.57° - 180° \simeq -108.43°$

αφού $\Theta \in (-\pi, \pi] = (-180, 180]$ (✓)

(β) Είναι $\vec{\beta} = (x_2, y_2) = (-1, 2)$ άρα $|\vec{\beta}| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$ μον.

και $\Theta_2 = \tan^{-1}\left(\frac{y_2}{x_2}\right) = \tan^{-1}\left(\frac{2}{-1}\right) = \tan^{-1}(-2) \simeq -63.43° + 180° \simeq 116.57°$ (✓)

(γ) Είναι $\vec{\alpha} + \vec{\beta} = -2\hat{i} - 6\hat{j} + (-\hat{i} + 2\hat{j}) = -2\hat{i} - \hat{i} - 6\hat{j} + 2\hat{j} = (-2-1)\hat{i} + (-6+2)\hat{j} = -3\hat{i} - 4\hat{j}$

άρα $\vec{\alpha}+\vec{\beta} = (x_3, y_3) = (-3, -4)$ και $|\vec{\alpha}+\vec{\beta}| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25} = 5$ μον.

~~οπότε~~ και $\Theta_3 = \tan^{-1}\left(\frac{y_3}{x_3}\right) = \tan^{-1}\left(\frac{-4}{-3}\right) = \tan^{-1}\left(\frac{4}{3}\right) \simeq 53.13° - 180° \simeq -126.87°$ (✓)

(δ) Είναι $\vec{\beta} - \vec{\alpha} = -\hat{i} + 2\hat{j} - (-2\hat{i} - 6\hat{j}) = -\hat{i} + 2\hat{i} + 2\hat{j} + 6\hat{j} = \hat{i} + 8\hat{j}$

άρα $\vec{\beta} - \vec{\alpha} = (x_4, y_4) = (1, 8)$ οπότε $|\vec{\beta} - \vec{\alpha}| = \sqrt{1^2 + 8^2} = \sqrt{1+64} = \sqrt{65}$ μον.

και $\Theta_4 = \tan^{-1}\left(\frac{8}{1}\right) = \tan^{-1}(8) \simeq 82.87°$ (✓)

(ε) Είναι $\vec{\alpha} - \vec{\beta} = -2\hat{i} - 6\hat{j} - (-\hat{i} + 2\hat{j}) = -2\hat{i} + \hat{i} - 6\hat{j} - 2\hat{j} = -\hat{i} - 8\hat{j}$

άρα $\vec{\alpha} - \vec{\beta} = (x_5, y_5) = (-1, -8) = -(\vec{\beta} - \vec{\alpha})$

οπότε $|\vec{\alpha} - \vec{\beta}| = +|\vec{\beta} - \vec{\alpha}| = +\sqrt{65}$ μον. και $\Theta_5 = +\Theta_4 - 180° = -180° + 82.87° = -97.13°$ (✓)

ή $|\vec{\alpha} - \vec{\beta}| = \sqrt{(-1)^2 + (-8)^2} = \sqrt{1+64} = \sqrt{65}$ και $\Theta_5 = \tan^{-1}\left(\frac{-8}{-1}\right) = \tan^{-1}(8) \simeq 82.87° - 180° = -\,''-$

(ζ) Ξέρουμε $\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos(\Theta)$ (→ συν(Θ)) άρα $\Rightarrow (\vec{\beta} - \vec{\alpha}) \cdot (\vec{\alpha} - \vec{\beta}) = |\vec{\beta} - \vec{\alpha}| \cdot |\vec{\alpha} - \vec{\beta}| \cdot \cos(\Theta)$

$\Rightarrow$ από προηγούμενα ερωτήματα είναι

$\Rightarrow (1+8)(-1-8) = (\sqrt{65})(\sqrt{65}) \cdot \cos(\Theta)$

$\Rightarrow (1)(-1) + (8)(-8) = (\sqrt{65})^2 \cdot \cos(\Theta)$

$\Rightarrow (-1) + (-64) = 65\cos(\Theta)$

$\Rightarrow -65 = 65\cos(\Theta)$ άρα $\cos(\Theta) = -\frac{65}{65} = -1$

και ισχύει για $\Theta = 180°$

Άσκηση 2: Έχουμε $\vec{\delta} = \vec{i} - \vec{j}$ και θέλουμε $\vec{x} = -\vec{\delta} = -\vec{i} + \vec{j}$

άρα $\vec{x} = -\vec{i} + \vec{j}$ (δείχνει αντίθετη κατεύθυνση) ή αντίθετη διεύθυνση.

όπως πρέπει $|\vec{x}| = 3$ άρα $|\vec{x}| = 3 \Leftrightarrow \sqrt{x^2+y^2} = 3$ αφού $\vec{x} = (x, y) = (-1, 1)$

οπότε $\sqrt{x^2+y^2} = 3 \overset{\text{2ο}}{\Rightarrow} x^2+y^2 = 9 \overset{*}{\Rightarrow} x^2+(-x)^2 = 9 \Rightarrow x^2+x^2 = 9 \Rightarrow 2x^2 = 9 \Rightarrow x^2 = \frac{9}{2} \Rightarrow \dots$

* $x = -1$ και $y = 1$ οπότε $y = -x$!

$\Rightarrow$ έτσι: $x = \sqrt{\frac{9}{2}} \Rightarrow x = \frac{\sqrt{9}}{\sqrt{2}} \Rightarrow x = \frac{3}{\sqrt{2}}$ και $x = \frac{3 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{3\sqrt{2}}{2}$

άρα $x = \frac{3\sqrt{2}}{2}$ και $y = -x = -\frac{3\sqrt{2}}{2}$ ή $\vec{x} = \left(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right) \rightarrow \vec{x} = \left(\frac{3\sqrt{2}}{2}\vec{i} - \frac{3\sqrt{2}}{2}\vec{j}\right)$

Άσκηση 3 Έχουμε $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$, $\vec{b} = -\vec{i} - 3\vec{j} + 2\vec{k}$ και $\vec{c} = 3\vec{i} + \vec{j} - \vec{k}$.

(α) $\vec{a} \cdot (\vec{b} + \vec{c}) = (\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (-\vec{i} - 3\vec{j} + 2\vec{k} + 3\vec{i} + \vec{j} - \vec{k}) =$

$= (\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (2\vec{i} - 2\vec{j} + \vec{k}) =$

$= (1 \cdot 2) + (2 \cdot (-2)) + ((-3) \cdot 1) =$

$= 2 + (-4) + (-3) = \underline{-5}$

(β) $\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = (\vec{i} + 2\vec{j} - 3\vec{k})(-\vec{i} - 3\vec{j} + 2\vec{k}) - (\vec{i} + 2\vec{j} - 3\vec{k})(3\vec{i} + \vec{j} - \vec{k})$

$= (\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (-\vec{i} - 3\vec{j} + 2\vec{k} - (3\vec{i} + \vec{j} - \vec{k}))$

$= (\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (-\vec{i} - 3\vec{j} + 2\vec{k} - 3\vec{i} - \vec{j} + \vec{k})$

$= (\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (-4\vec{i} - 4\vec{j} + 3\vec{k})$

$= (1 \cdot (-4)) + (2 \cdot (-4)) + ((-3) \cdot 3)$

$= -4 - 8 - 9 = \underline{-21}$

(γ) $(\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{c}) = [(\vec{i} + 2\vec{j} - 3\vec{k})(3\vec{i} + \vec{j} - \vec{k})] [(-\vec{i} - 3\vec{j} + 2\vec{k})(3\vec{i} + \vec{j} - \vec{k})]$

$= [(1 \cdot 3) + (2 \cdot 1) + ((-3)(-1))] [((-1)(3)) + ((-3) \cdot 1) + (2(-1))]$

$= [8][-8] = \underline{-64}$

Άσκηση 4 Έχουμε $\vec{a} = 2\vec{i} - \vec{j}$ και $\vec{b} = \vec{i} + 2\vec{j}$ και $\vec{c} \perp \vec{a}$, $\vec{c} \cdot \vec{b} = 1$.

- Αφού $\vec{c} \perp \vec{a}$ ισχύει $\vec{c} \cdot \vec{a} = 0 \Rightarrow$ Έστω $\vec{c} = (x, y)$ τότε $(x\vec{i} + y\vec{j})(2\vec{i} - \vec{j}) = 0 \Rightarrow$

$\Rightarrow (x \cdot 2) + (y(-1)) = 0 \Rightarrow 2x - y = 0$ (1)

- Αφού $\vec{c} \cdot \vec{b} = 1 \Rightarrow (x\vec{i} + y\vec{j})(\vec{i} + 2\vec{j}) = 1 \Rightarrow (x \cdot 1) + (y \cdot 2) = 1 \Rightarrow x + 2y = 1$ (2)

- Είναι $\begin{cases} (1)\ 2x - y = 0 \\ (2)\ x + 2y = 1 \end{cases} \overset{(-2)}{\Rightarrow} \begin{cases} 2x - y = 0 \\ -2x - 4y = -2 \end{cases} \overset{(+)}{\Rightarrow} 2x - 2x - y - 4y = 0 - 2 \Rightarrow -5y = -2 \Rightarrow \underline{y = \frac{2}{5}}$

και για $y = \frac{2}{5}$: (1) $2x - \frac{2}{5} = 0 \Rightarrow 2x = \frac{2}{5} \Rightarrow x = \frac{\frac{2}{5}}{2} \Rightarrow x = \frac{2}{10} = \frac{1}{5}$

οπότε $\vec{c} = x\vec{i} + y\vec{j} = \underline{\frac{1}{5}\vec{i} + \frac{2}{5}\vec{j} = \vec{c}}$

Άσκηση 5:

(α) $y(x) = \cos(\ln(x)) = συν(\ln x)$

· $y^{\circ}(x) = \cos(u)$ αν $u = \ln x$

· $y'(x) = (\cos(u))' = \cos'(u) \cdot u' = -\sin(u) \cdot (\ln x)' = -\sin(u) \cdot \frac{1}{x} = -\frac{\sin(\ln(x))}{x}$

(β) $y(x) = (2x+1)^{-3}$

· Έστω $u = 2x+1$ τότε $y(x) = u^{-3}$

· άρα $y'(x) = (u^{-3})' = -3u^{-4} \cdot u' = -3u^{-4}(2x+1)' = -3u^{-4}(2 \cdot 1 + 0) = -6u^{-4}$

· έτσι $y'(x) = -6(2x+1)^{-4}$

(γ) $y(x) = (5x^3 - x^4)^7$

· Αν $u = 5x^3 - x^4$ τότε $y(x) = u^7$

· άρα $y'(x) = 7u^6 \cdot u' = 7u^6(3 \cdot 5x^{3-1} - 4x^{4-1}) = 7u^6(15x^2 - 4x^3)$

· οπότε $y'(x) = 7(5x^3 - x^4)^7 \cdot (15x^2 - 4x^3)$

(δ) $y(x) = \frac{x^2-1}{x^2+2}$

· Είναι $y'(x) = \frac{(x^2-1)'(x^2+2) - (x^2-1)(x^2+2)'}{((x^2+2))^2} = \frac{(2x)(x^2+2) - (x^2-1)(2x)}{(x^2+2)^2} =$

$= \frac{2x((x^2+2) - (x^2-1))}{(x^2+2)^2} = \frac{2x(3)}{(x^2+2)^2} = \frac{6x}{(x^2+2)^2}$

Άσκηση 6:

(α) $\int_1^4 \left(\frac{3}{2}\sqrt{x} - \frac{4}{x^2}\right) dx = \underbrace{\int_1^4 \frac{3}{2}\sqrt{x}\, dx} + \underbrace{\int_1^4 -\frac{4}{x^2} dx} = I_1 + I_2 = *$

$* I_2 = \int_1^4 -4 \cdot \frac{1}{x^2} dx = 4\int_1^4 -\frac{1}{x^2} dx = 4\left[\frac{1}{x}\right]_1^4 = 4\left(\frac{1}{4} - \frac{1}{1}\right) = 4\left(\frac{1}{4} - \frac{4}{4}\right) = 4\left(-\frac{3}{4}\right) = \underline{-3}$

$* I_1 = \int_1^4 \frac{3}{2}\sqrt{x}\, dx = \frac{3}{2}\int_1^4 \sqrt{x}\, dx = \frac{3}{2}\int_1^4 x^{\frac{1}{2}} dx = \frac{3}{2}\left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}\right]_1^4 = \frac{3}{2}\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^4 =$

$= \frac{3}{2}\left[\frac{2\sqrt{x^3}}{3}\right]_1^4 = \frac{3}{2}\left[\frac{2}{3}\sqrt{x^3}\right]_1^4 = \left[\sqrt{x^3}\right]_1^4 = \sqrt{4^3} - \sqrt{1^3} = \sqrt{64} - \sqrt{1} = 8 - 1 = \underline{7}$

άρα $I_1 + I_2 = 7 + (-3) = \underline{4}$

→

(β) $\int_0^{\frac{\pi}{6}} (1-\cos(3x))\cdot(\sin(3x))\,dx =$

$= \int_0^{\frac{\pi}{6}} (\sin(3x) - \cos(3x)\sin(3x))\,dx =$

$= \int_0^{\frac{\pi}{6}} \sin(3x)\,dx - \int_0^{\frac{\pi}{6}} \cos(3x)\sin(3x)\,dx = I_1 - I_2 = *$

* $I_1 = \int_0^{\frac{\pi}{6}} \sin(3x)\,dx = \left[-\frac{1}{3}\cos(3x)\right]_0^{\frac{\pi}{6}} = -\frac{1}{3}\left[\cos(3x)\right]_0^{\frac{\pi}{6}} = -\frac{1}{3}\left(\cos(3\tfrac{\pi}{6}) - \cos(3\cdot 0)\right) =$

$= -\frac{1}{3}\left(\cos(\tfrac{\pi}{2}) - \cos(0)\right) = -\frac{1}{3}(0-1) = \underline{\frac{1}{3}}$

* $I_2 = \int_0^{\frac{\pi}{6}} \cos(3x)\sin(3x)\,dx = \int_0^{\frac{\pi}{6}} \frac{1}{2}\cdot\sin(2\cdot 3x)\,dx = \int_0^{\frac{\pi}{6}} \frac{1}{2}\cdot\sin(6x)\,dx = \frac{1}{2}\int_0^{\frac{\pi}{6}} \sin(6x)\,dx =$

$= \frac{1}{2}\left[-\frac{1}{6}\cos(6x)\right]_0^{\frac{\pi}{6}} = -\frac{1}{12}\left[\cos(6x)\right]_0^{\frac{\pi}{6}} = -\frac{1}{12}\left(\cos(\pi) - \cos(0)\right) =$

$= -\frac{1}{12}(-1-1) = -\frac{1}{12}(-2) = \frac{2}{12} = \underline{\frac{1}{6}}$

άρα $I_1 - I_2 = \frac{1}{3} - \frac{1}{6} = \frac{2}{6} - \frac{1}{6} = \underline{\frac{1}{6}}$

ΤΕΛΟΣ

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